HARDWARE ENGINEER GUIDE

AI-Based Monitoring Solution for Oil & Gas Industry

Document Version: 1.0 **Date**: December 2025 **Project Code**: OG-AI-MON-2025 **Audience**: Hardware Engineers, Embedded Systems Engineers, Field Technicians

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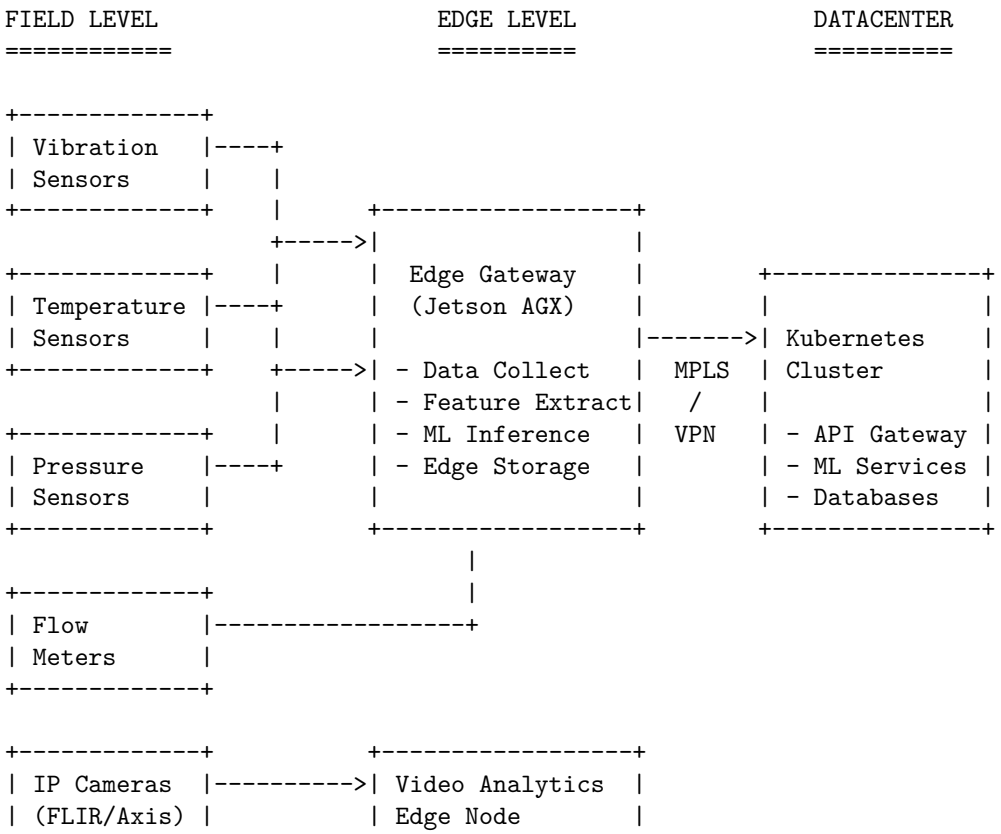
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1. SYSTEM HARDWARE OVERVIEW

1.1 Architecture Diagram



+-------------+ +------------------+

1.2 Bill of Materials Summary

Category	Item	Qty	Unit Cost	Total
Sensors				**$304,000**
	Vibration Sensor Kit	200	$850	$170,000
	Temperature Sensor	200	$120	$24,000
	Pressure Transmitter	150	$450	$67,500
	Flow Meter	50	$850	$42,500
Edge Computing				**$360,000**
	Jetson AGX Orin 64GB	20	$2,199	$43,980
	Industrial Edge PC	20	$1,500	$30,000
	Edge Enclosure	20	$800	$16,000
	UPS (Edge)	20	$600	$12,000
	Installation Kit	20	$400	$8,000
Cameras				**$540,000**
	Axis P1468-LE	30	$1,200	$36,000
	FLIR A70 Thermal	30	$4,500	$135,000
	GigE Vision Cables	60	$50	$3,000
	Camera Housing (ATEX)	30	$2,200	$66,000
Servers				**$320,000**
	Dell R750xa (2xA100)	4	$65,000	$260,000
	Dell R650 (CPU)	6	$10,000	$60,000
TOTAL				**$1,524,000**

1.3 Hardware Selection Criteria

Requirement	Specification	Rationale
ATEX/IECEx	Zone 1 rated	Hazardous area operation
Operating Temp	-40°C to +70°C	Outdoor/desert environments
IP Rating	IP66 minimum	Dust/water protection
EMC	IEC 61326-1	Industrial immunity
Power	24VDC nominal	Standard industrial supply
Communication	Modbus RTU/TCP, MQTT	Legacy + modern protocols
MTBF	>50,000 hours	Reliability requirement

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2. SENSOR SYSTEMS

2.1 Vibration Sensors

Primary Specification

Model: PCB Piezotronics 682C09 (or equivalent)

Parameter	Specification
Type	IEPE Accelerometer
Sensitivity	100 mV/g

Parameter	Specification
Frequency Range	0.5 Hz to 14 kHz
Dynamic Range	80 g peak
Temperature Range	-54°C to +121°C
Output	4-20mA + HART
Housing	316L Stainless Steel
Mounting	1/4-28 UNF stud
Certification	ATEX/IECEx Zone 1

Mounting Guidelines

```
VIBRATION SENSOR MOUNTING
=========================

MOTOR BEARING MONITORING:
                                    +---------+
    +----------------------------+  | Sensor  |
    |                            |  +---------+
    |    +-------+  +-------+    |      |
    |    |       |  |       |    |      V
    |    | DE    |  | NDE   |    |  Mount at 45deg
    |    | Brg   |  | Brg   |    |  to shaft axis
    |    +---+---+  +---+---+    |
    |        |          |        |
    |========|==========|=======|  Shaft
    |                            |
    +----------------------------+

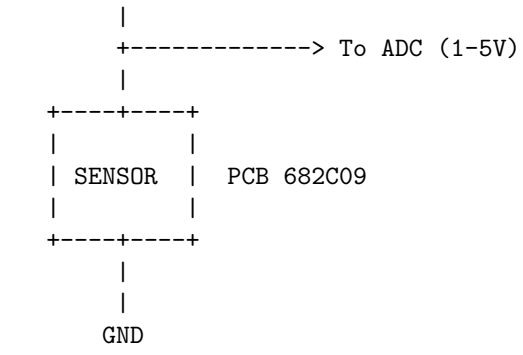
Mounting Positions:
- Horizontal (H): Radial, 3 o'clock
- Vertical (V): Radial, 12 o'clock
- Axial (A): Along shaft centerline

Surface Preparation:
1. Clean mounting surface with isopropyl alcohol
2. Ensure flatness <0.001" per inch
3. Apply thin layer of silicone grease
4. Torque stud to 1.5 Nm (13 in-lb)
```

Wiring Diagram

```
VIBRATION SENSOR WIRING (4-20mA + HART)
=======================================

              +24VDC
                |
                |
           +----+----+
           |         |
           |  250R   |  Precision resistor
           |  0.1%   |  for 1-5V output
           |         |
           +----+----+
```




```
Cable: Belden 8719 (shielded twisted pair)
Max Length: 300m
Shield: Ground at junction box only
```

Sensor Data Acquisition

Parameter	Value
Sampling Rate	25.6 kHz
Samples per Acquisition	8192
Acquisition Interval	10 minutes
Data Format	16-bit signed
Anti-aliasing Filter	10 kHz (4th order Butterworth)

2.2 Temperature Sensors

Primary Specification

Model: Emerson 644H with RTD

Parameter	Specification
Element	Pt100 RTD, Class A
Range	-200°C to +600°C
Accuracy	±0.1°C
Output	4-20mA + HART
Response Time	T90 < 3s (in water)
Housing	316SS with thermowell
Process Connection	1/2” NPT
Certification	ATEX/IECEx Zone 1

Thermowell Selection

```
THERMOWELL SIZING
=================

Wake Frequency Check (API 2566):

        fs   >  1.25
        fn

Where:
```

```
- fs = vortex shedding frequency = (St × V) / d
- fn = natural frequency of thermowell
- St = Strouhal number (~0.22)
- V = flow velocity (m/s)
- d = tip diameter (m)

Example:
- Pipe: 6" NPS
- Flow: 3 m/s
- Thermowell: 6.35mm OD

fs = (0.22 × 3) / 0.00635 = 104 Hz
fn (standard) = 250 Hz
Ratio = 250/104 = 2.4 > 1.25  [PASS]
```

2.3 Pressure Transmitters

Primary Specification

Model: Emerson 3051S

Parameter	Specification
Range	0-100 bar (configurable)
Accuracy	±0.025% of span
Output	4-20mA + HART
Process Connection	1/2” NPT
Wetted Materials	316SS diaphragm
Response Time	100ms
Certification	ATEX/IECEx Zone 1, SIL 2

Installation Requirements

```
PRESSURE TRANSMITTER INSTALLATION
================================

LIQUID SERVICE:
                    Process Pipe
    ================|================
                    |
                    | Impulse line
                    | (slope down)
                    |
               +----+----+
               | Isolator|
               | Valve   |
               +----+----+
                    |
               +----+----+
               |   PT    |
               |  3051S  |
               +----+----+

- Mount transmitter BELOW tapping point
- Slope impulse line 1:12 minimum
```

```
- Use impulse line <15m length
- Vent at high point for gas service


GAS SERVICE:
                    Process Pipe
    ================|================
                     |
                +----+----+
                |   PT    |  Mount ABOVE
                |  3051S  |  tapping point
                +----+----+
                     |
                     | Impulse line
                     | (slope up)
                     |
                Condensate pot
                (if required)
```

2.4 Flow Meters

Primary Specification

Model: Emerson Micro Motion F-Series

Parameter	Specification
Type	Coriolis mass flow
Range	0-50 kg/min
Accuracy	±0.1% of rate
Output	4-20mA + HART + Modbus
Process Connection	2” ANSI 300# RF
Materials	316L SS
Certification	ATEX/IECEx Zone 1, Custody Transfer

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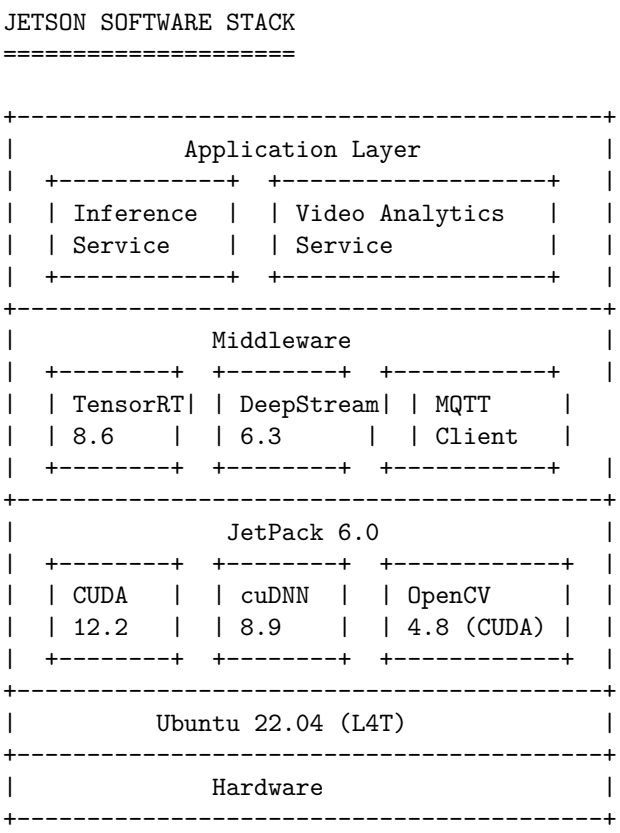
3. EDGE COMPUTING DEVICES

3.1 NVIDIA Jetson AGX Orin 64GB

Specifications

Parameter	Specification
GPU	2048 CUDA cores, 64 Tensor cores
CPU	12-core Arm Cortex-A78AE
Memory	64GB LPDDR5
Storage	64GB eMMC + NVMe slot
AI Performance	275 TOPS (INT8)
Power	15W - 60W (configurable)
Interfaces	GbE, USB 3.2, PCIe, GPIO
Operating Temp	-25°C to +80°C

Software Stack



Power Configuration

```
# Set power mode via nvpmodel
# Mode 0: MAXN (60W) - Maximum performance
# Mode 1: 30W - Balanced
# Mode 2: 15W - Power saving

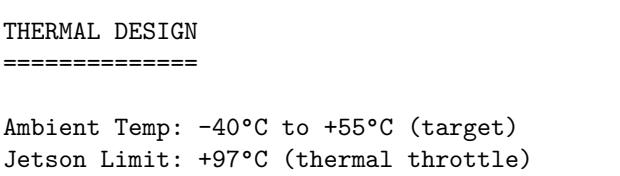
# Check current mode
nvpmodel -q

# Set to 30W mode
sudo nvpmodel -m 1

# Set to maximum performance
sudo nvpmodel -m 0

# Enable all CPU cores
sudo jetson_clocks
```

Thermal Management



```
Design Target: <85°C junction temp

Cooling Solution:
+---------------------------+
|   Active Heatsink         |
|   +-------------------+   |
|   |  80mm Fan         |   |  50 CFM @ 5000 RPM
|   +-------------------+   |
|   |  Aluminum Fins    |   |  Rth = 0.3 K/W
|   +-------------------+   |
|   |  Copper Base      |   |  Thermal paste: 5 W/mK
|   +-------------------+   |
|   |  Jetson Module    |   |
|   +-------------------+   |
+---------------------------+

Thermal Budget (60W mode, 55°C ambient):
- Junction temp = Ambient + (Power × Rth)
- Junction temp = 55 + (60 × 0.3) = 73°C < 85°C [OK]
```

3.2 Industrial Edge Gateway

Specifications (Advantech UNO-2484G)

Parameter	Specification
CPU	Intel Core i7-1185G7E
Memory	32GB DDR4
Storage	512GB NVMe SSD
Expansion	2x PCIe, 2x Mini-PCIe
I/O	4x GbE, 4x USB, 4x COM
Power	24VDC (10-30V range)
Operating Temp	-20°C to +60°C
Certification	UL, CE, FCC

I/O Configuration

```
EDGE GATEWAY I/O MAPPING
========================

COM Ports:
- COM1 (RS-485): Modbus RTU sensors
- COM2 (RS-485): Modbus RTU sensors
- COM3 (RS-232): Debug console
- COM4 (RS-232): Reserved

Ethernet Ports:
- ETH1: Field network (sensors, PLCs)
- ETH2: Uplink to datacenter
- ETH3: Camera network
- ETH4: Management/IPMI

GPIO (via PCIe expansion):
- DI 1-8: Digital inputs (24VDC)
```

```
- DO 1-4: Digital outputs (relay)
- AI 1-4: 4-20mA analog inputs
```

3.3 Edge Device Comparison

Feature	Jetson AGX	Advantech UNO	Selection Criteria
Use Case	AI Inference	Data Gateway	
GPU	2048 CUDA	None	ML inference
AI Performance	275 TOPS	N/A	Video analytics
Industrial I/O	Limited	Extensive	Sensor connectivity
Protocol Support	Limited	Modbus, OPC-UA	Legacy systems
Operating Temp	-25 to +80°C	-20 to +60°C	
Power	15-60W	35W	
Cost	$2,199	$1,500	

Recommendation: - Use Jetson AGX for: Video analytics, ML inference - Use Advantech UNO for: Data aggregation, protocol conversion

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4. CAMERA SYSTEMS

4.1 Visible Light Cameras

Primary Specification (Axis P1468-LE)

Parameter	Specification
Resolution	8MP (3840x2160)
Frame Rate	30 fps @ 4K
Lens	4.3-8.6mm varifocal
Low Light	0.03 lux (F1.0)
Compression	H.265, H.264, MJPEG
Interface	GigE (PoE+)
Operating Temp	-40°C to +60°C
IP Rating	IP66/IP67
Certification	NEMA 4X

Camera Placement for PPE Detection

```
PPE DETECTION CAMERA PLACEMENT
==============================

Entry Point Monitoring:

        +-----------+
        |  Door     |
        |           |
        +-----------+
              |
              | 3-5m distance
              |
```

```
        V
      +-----+-----+
      |  Camera   |  Height: 2.5-3m
      |  Axis     |  Angle: 15-30 deg down
      |  P1468-LE |  FOV: Capture full body
      +-----------+

Field of View Calculation:
- Sensor: 1/1.8" (7.1mm × 5.3mm)
- Lens: 4.3mm
- Horizontal FOV = 2 × arctan(7.1 / (2 × 4.3)) = 79°
- At 5m: Coverage width = 2 × 5 × tan(79°/2) = 7.8m
```

4.2 Thermal Cameras

Primary Specification (FLIR A70)

Parameter	Specification
Resolution	640×480
Thermal Sensitivity	<30mK (NETD)
Temperature Range	-20°C to +650°C
Accuracy	±2°C or ±2%
Frame Rate	30 Hz
Lens	24° × 18° FOV
Interface	GigE Vision
Operating Temp	-25°C to +50°C
Certification	ATEX Zone 2 (with housing)

Thermal Camera Applications

Application	Configuration	Alarm Threshold
Hot spot detection	Relative mode	+15°C above ambient
Gas leak imaging	OGI filter	N/A (visual)
Electrical fault	Spot meter	>70°C
Bearing overheat	ROI monitoring	>80°C
Fire detection	Rate of rise	>10°C/s

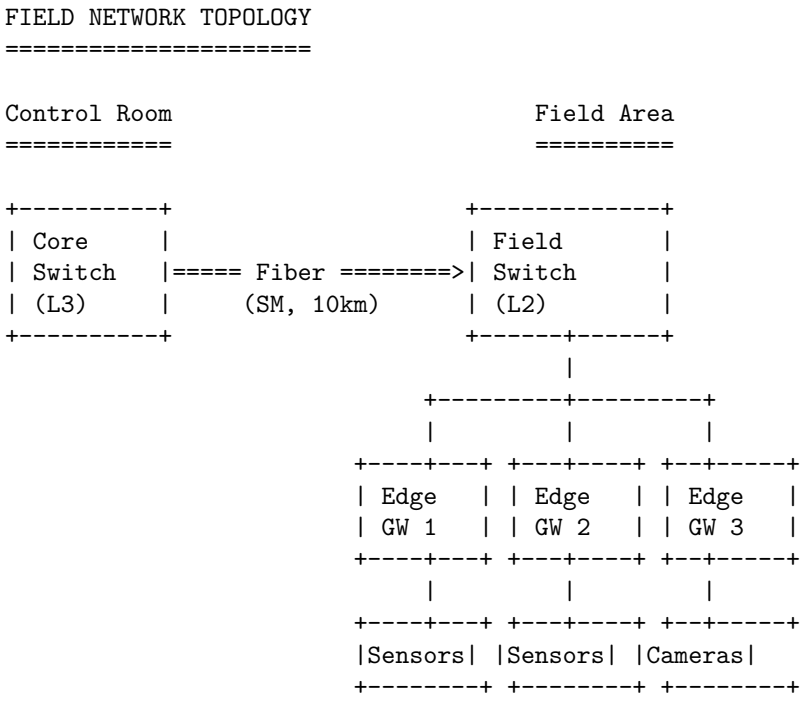
4.3 ATEX Camera Housing

Specification (Axis XF40-Q2901)

Parameter	Specification
Material	316L Stainless Steel
Window	Germanium (thermal), Glass (visible)
Certification	ATEX Zone 1, IECEx
IP Rating	IP66/IP68
Operating Temp	-40°C to +65°C
Pressurization	Px purge available

5. NETWORK INFRASTRUCTURE

5.1 Field Network Architecture



5.2 Industrial Ethernet Switch

Specification (Cisco IE-4010)

Parameter	Specification
Ports	24x GbE + 4x 10G SFP+
PoE	30W per port (PoE+)
Redundancy	REP, MRP, PRP/HSR
Operating Temp	-40°C to +75°C
MTBF	500,000 hours
Certification	Class 1 Div 2, ATEX Zone 2

Network Segmentation

VLAN	Name	Purpose	Subnet
10	Management	Switch management	10.10.10.0/24
20	Sensors	Field sensors	10.10.20.0/24
30	Cameras	Video streams	10.10.30.0/24
40	Edge	Edge devices	10.10.40.0/24
50	Uplink	Datacenter link	10.10.50.0/24

5.3 Wireless Options

Industrial WiFi (Cisco Aironet IW6300H)

Parameter	Specification
Standard	WiFi 6 (802.11ax)
Frequency	2.4/5/6 GHz
Data Rate	Up to 2.4 Gbps
Range	150m (outdoor)
Operating Temp	-40°C to +70°C
Certification	Hazloc, IP67

LoRaWAN (for remote sensors)

Parameter	Specification
Frequency	915 MHz (US)
Range	Up to 15km (LOS)
Data Rate	0.3 - 50 kbps
Battery Life	10+ years
Use Case	Remote well monitoring

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6. POWER SYSTEMS

6.1 Power Distribution

```
POWER DISTRIBUTION ARCHITECTURE
================================

Main Panel (480VAC 3-phase)
           |
           V
    +------+------+
    |  UPS 10kVA  |
    |  (Online)   |
    +------+------+
           |
           V
    +------+------+
    | Distribution|  120/240VAC
    |    Panel    |
    +------+------+
           |
    +------+------+------+
    |      |      |      |
    V      V      V      V
+------+ +------+ +------+ +------+
| 24V  | | 24V  | | 48V  | | PoE  |
| PSU  | | PSU  | | PSU  | | Inj  |
| 10A  | | 10A  | | 5A   | | 30W  |
+------+ +------+ +------+ +------+
   |        |        |        |
Sensors  Sensors  Edge    Cameras
                  Nodes
```

6.2 Power Supply Selection

24VDC Power Supply

Model: Mean Well NDR-240-24

Parameter	Specification
Output	24VDC, 10A
Input	90-264VAC
Efficiency	93%
Operating Temp	-30°C to +70°C
Mounting	DIN rail
Protection	OVP, OCP, SCP
MTBF	536,000 hours

Power Budget Calculation

```
POWER BUDGET - EDGE NODE CABINET
================================

Item                          Qty    Power(W)   Total(W)
-------------------------------------------------------
Jetson AGX Orin               1      60         60
Advantech UNO-2484G           1      35         35
Industrial Switch             1      25         25
Vibration DAQ                 2      15         30
Misc (fans, LEDs)             -      10         10
-------------------------------------------------------
                              SUBTOTAL:         160W
                              Safety (25%):      40W
                              TOTAL:            200W

Power Supply Selection:
- Required: 200W @ 24V = 8.3A
- Selected: 24V/10A = 240W
- Margin: 40W (20% headroom) [OK]
```

6.3 UPS System

Edge UPS Specification

Model: APC Smart-UPS SRT 1500VA

Parameter	Specification
Capacity	1500VA / 1350W
Input	120VAC
Output	120VAC, pure sine
Runtime	7 min @ full load
Battery	Lead-acid (replaceable)
Interface	Network card, dry contacts
Operating Temp	0°C to +40°C

Battery Runtime Calculation

```
BATTERY RUNTIME CALCULATION
```

```
==========================

Load: 200W
UPS: 1500VA, 1350W
Battery: 2 × 12V, 9Ah = 216Wh

Efficiency: 85%
Available energy = 216 × 0.85 = 184Wh

Runtime = 184Wh / 200W = 0.92 hours = 55 minutes

Note: Manufacturer states 7 min at full load (1350W)
At 200W (~15% load), expect ~45-60 min runtime
```

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7. ENCLOSURES & ENVIRONMENTAL

7.1 ATEX/IECEx Zones

```
HAZARDOUS AREA CLASSIFICATION
============================

Zone 0: Explosive atmosphere continuously present
        - Sensors only (IS barriers required)

Zone 1: Explosive atmosphere likely during normal operation
        - Sensors with Ex d or Ex e protection
        - Cameras in pressurized housings

Zone 2: Explosive atmosphere unlikely
        - Edge devices in suitable enclosures
        - Standard IP66 equipment acceptable

Safe Area: No explosive atmosphere expected
        - Standard equipment
        - Control room, datacenter
```

7.2 Enclosure Specifications

Field Cabinet (Zone 2)

Model: Rittal CS 9783.500

Parameter	Specification
Dimensions	800W × 2000H × 400D mm
Material	Powder-coated steel
IP Rating	IP66
Certification	ATEX Zone 2
Climate Control	500W cooling unit
Temp Range	-40°C to +50°C internal

Intrinsic Safety Barriers

Model: Pepperl+Fuchs KFD2-STC4-Ex1

Parameter	Specification
Input	4-20mA from field
Output	4-20mA to safe area
Isolation	2.5kV
Max Voltage	28VDC
Ex Rating	[Ex ia] IIC
SIL	SIL 2

```
IS BARRIER CONNECTION
=====================

HAZARDOUS AREA           |    SAFE AREA
(Zone 1)                 |
                         |
+------------+      +----+----+      +----------+
| Field      |------|  IS     |------| Edge     |
| Sensor     |  Blue|  Barrier|      | Gateway  |
| (4-20mA)   |  Wire|  KFD2   |      | (ADC)    |
+------------+      +----+----+      +----------+
                         |
                         |
              Max 100m   |   Max 300m

Blue Wire: IS-rated cable (capacitance <100nF/km)
```

7.3 Environmental Protection

IP66 Requirements

Test	Method	Requirement
Dust	IEC 60529	No ingress
Water	Powerful jets	No harmful effects
Temp	IEC 60068-2	-40°C to +70°C
Vibration	IEC 60068-2-6	10-150Hz, 2g
Shock	IEC 60068-2-27	30g, 11ms

Corrosion Protection

Environment	Protection Level	Treatment
Offshore	C5-M (Very High)	316L SS, epoxy coating
Desert	C4 (High)	Hot-dip galvanized + paint
Refinery	C4 (High)	Powder coat + sealant

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8. INSTALLATION PROCEDURES

8.1 Pre-Installation Checklist

```
PRE-INSTALLATION CHECKLIST
==========================

Site Survey:
[ ] Hazardous area classification confirmed
[ ] Power availability verified (voltage, phase, amps)
[ ] Network infrastructure reviewed
[ ] Mounting locations identified
[ ] Cable routing planned
[ ] Environmental conditions assessed
[ ] Permits obtained

Equipment:
[ ] All components received and inspected
[ ] Certificates verified (ATEX, calibration)
[ ] Firmware versions checked
[ ] Configuration files prepared
[ ] Spare parts available

Safety:
[ ] Work permit issued
[ ] Hot work permit (if required)
[ ] Confined space permit (if required)
[ ] LOTO procedure reviewed
[ ] PPE requirements confirmed
[ ] Emergency procedures briefed
```

8.2 Sensor Installation

Vibration Sensor Installation

```
VIBRATION SENSOR INSTALLATION PROCEDURE
=======================================

1. Surface Preparation
   a. Clean mounting area with solvent
   b. Check surface flatness (<0.001" per inch)
   c. Mark mounting location

2. Drill and Tap
   a. Center punch location
   b. Drill pilot hole (5.1mm for 1/4-28 UNF)
   c. Tap threads (1/4-28 UNF, 12mm depth)
   d. Clean threads with compressed air

3. Mount Sensor
   a. Apply thin layer of silicone grease to base
   b. Thread sensor into mounting hole
   c. Hand tighten, then use wrench
   d. Torque to 1.5 Nm (13 in-lb)
```

```
4. Cable Connection
   a. Route cable through conduit
   b. Connect to junction box
   c. Verify polarity (signal +, signal -, shield)
   d. Ground shield at junction box only

5. Verification
   a. Check resistance (should be 2-3 ohms)
   b. Measure bias voltage (should be 8-12VDC)
   c. Tap sensor, verify signal response
   d. Record installation details

Documentation Required:
- Asset tag
- Sensor serial number
- Mounting location (drawing reference)
- Cable tag number
- Calibration certificate
- Photo of installation
```

8.3 Edge Device Installation

```
EDGE DEVICE INSTALLATION PROCEDURE
==================================

1. Cabinet Preparation
   a. Verify cabinet is properly grounded
   b. Install DIN rails and cable management
   c. Install power distribution block
   d. Install network patch panel

2. Power Installation
   a. Install 24VDC power supply on DIN rail
   b. Connect AC input (verify voltage)
   c. Set output to 24.0VDC (adjust if needed)
   d. Install UPS and connect battery

3. Network Installation
   a. Install industrial switch
   b. Connect fiber uplink (verify link light)
   c. Patch cables to panel
   d. Label all ports

4. Edge Device Installation
   a. Mount Jetson AGX on mounting plate
   b. Connect power (24VDC via DC-DC converter)
   c. Connect network cables
   d. Connect sensor interfaces

5. Cable Management
   a. Separate power and signal cables
   b. Use appropriate cable ties
   c. Label all cables at both ends
   d. Install strain reliefs
```

```
6. Power-Up Sequence
   a. Verify all connections
   b. Turn on AC power
   c. Verify 24VDC output
   d. Power on switch (wait for boot)
   e. Power on edge devices
   f. Verify network connectivity

7. Configuration
   a. Access device via SSH/console
   b. Apply network configuration
   c. Deploy application software
   d. Verify sensor data acquisition
   e. Test MQTT connectivity to cloud
```

---

9. CALIBRATION & COMMISSIONING

9.1 Sensor Calibration

Vibration Sensor Calibration

```
VIBRATION SENSOR CALIBRATION
============================

Reference: ISO 16063-21

Equipment Required:
- Calibrated reference accelerometer
- Vibration exciter (shaker)
- Signal analyzer
- Calibration certificate traceable to NIST

Procedure:
1. Mount sensor on shaker (back-to-back with reference)
2. Apply sinusoidal excitation at 159.2 Hz (1000 rad/s)
3. Measure sensor output and reference output
4. Calculate sensitivity: S = V_out / a_ref
5. Compare to nominal sensitivity (100 mV/g ± 5%)
6. Record calibration data

Acceptance Criteria:
- Sensitivity: 95-105 mV/g
- Frequency response: ±3dB from 10Hz to 10kHz
- Cross-axis sensitivity: <5%

Calibration Interval: 12 months
```

Temperature Sensor Calibration

```
TEMPERATURE SENSOR CALIBRATION
==============================
```

```
Reference: IEC 60751

Method: Comparison with reference PRT in dry-block calibrator

Calibration Points:
| Point | Temperature | Tolerance |
|-------|-------------|-----------|
| 1     | 0°C         | ±0.3°C    |
| 2     | 50°C        | ±0.3°C    |
| 3     | 100°C       | ±0.35°C   |
| 4     | 150°C       | ±0.4°C    |
| 5     | 200°C       | ±0.45°C   |

Procedure:
1. Insert sensor in dry-block calibrator
2. Allow stabilization (15 min per point)
3. Record 10 readings at 10-second intervals
4. Calculate mean and standard deviation
5. Compare to reference value
6. Apply correction if required

Calibration Interval: 24 months
```

9.2 System Commissioning

Loop Check Procedure

```
LOOP CHECK PROCEDURE
====================

For each sensor-to-edge loop:

1. Field Verification
   [ ] Sensor tag matches P&ID
   [ ] Sensor range matches datasheet
   [ ] Cable routing per design
   [ ] Terminations tight and correct

2. Signal Injection
   [ ] Apply 4mA signal -> Verify 0% reading
   [ ] Apply 12mA signal -> Verify 50% reading
   [ ] Apply 20mA signal -> Verify 100% reading
   [ ] Record actual vs expected

3. Edge Device Verification
   [ ] Data appearing in edge gateway
   [ ] Engineering units correct
   [ ] Timestamp accurate (NTP synced)
   [ ] Data reaching cloud platform

4. Alarm Testing
   [ ] Trigger low alarm -> Verify alert
   [ ] Trigger high alarm -> Verify alert
   [ ] Clear alarm -> Verify reset
```

```
5. Documentation
   [ ] Complete loop check form
   [ ] Sign-off by field tech
   [ ] Sign-off by control room
   [ ] Update as-built drawings
```

SAT (Site Acceptance Test)

```
SITE ACCEPTANCE TEST CHECKLIST
==============================

1. Infrastructure
   [ ] All equipment installed per design
   [ ] Power systems operational
   [ ] Network connectivity verified
   [ ] Environmental controls functional

2. Data Acquisition
   [ ] All sensors reading correctly
   [ ] Data rates as specified
   [ ] No data gaps >10 seconds
   [ ] Historical data accessible

3. ML Inference
   [ ] Models loaded on edge devices
   [ ] Inference latency <100ms
   [ ] Anomaly detection working
   [ ] Predictions matching expected

4. Alerts
   [ ] Critical alerts within 5 seconds
   [ ] Email notifications received
   [ ] SMS notifications received
   [ ] Dashboard updated

5. Performance
   [ ] CPU utilization <70%
   [ ] Memory utilization <80%
   [ ] Network bandwidth <50%
   [ ] Storage growth as expected

6. Documentation
   [ ] As-built drawings complete
   [ ] O&M manuals delivered
   [ ] Training completed
   [ ] Warranty registered
```

---

10. MAINTENANCE & TROUBLESHOOTING

10.1 Preventive Maintenance Schedule

Component	Task	Frequency	Duration
Sensors	Visual inspection	Monthly	5 min/sensor
Sensors	Calibration check	Annually	30 min/sensor
Edge devices	Firmware update	Quarterly	1 hour
Edge devices	Log review	Weekly	15 min
Cameras	Lens cleaning	Monthly	10 min
Cameras	Focus verification	Quarterly	20 min
Enclosures	Seal inspection	Quarterly	10 min
Enclosures	Filter replacement	Quarterly	15 min
Network	Cable inspection	Annually	2 hours
UPS	Battery test	Monthly	5 min
UPS	Battery replacement	3-5 years	1 hour

10.2 Common Issues and Solutions

Vibration Sensor Issues

Symptom	Possible Cause	Solution
No signal	Cable break	Check continuity
No signal	Power issue	Verify 24VDC at sensor
Noisy signal	Ground loop	Check shield grounding
Noisy signal	EMI	Re-route cables
Drift	Loose mounting	Re-torque sensor
Drift	Temperature effect	Check temp compensation
Saturation	Overrange	Verify sensor range

Edge Device Issues

Symptom	Possible Cause	Solution
No boot	Power failure	Check 24VDC, fuses
No boot	Corrupted OS	Re-flash from backup
High CPU	Runaway process	Restart service
High temp	Fan failure	Replace fan
High temp	Blocked airflow	Clean filters
No network	Cable/switch issue	Check LEDs, swap cable
No cloud connect	VPN issue	Restart VPN client

Diagnostic Commands

```
# Edge device diagnostics

# Check system status
systemctl status inference-service
systemctl status mqtt-client

# Check sensor data acquisition
journalctl -u data-acquisition -f

# Check network connectivity
ping -c 5 gateway.oilgas.internal
```

```
curl -s http://localhost:8080/health | jq

# Check GPU status (Jetson)
tegrastats
nvidia-smi

# Check disk usage
df -h
du -sh /var/log/*

# Check memory
free -h
top -bn1 | head -20

# Check temperatures
cat /sys/class/thermal/thermal_zone*/temp

# Test sensor connection
python3 -c "import serial; s=serial.Serial('/dev/ttyUSB0', 9600); print(s.read(100))"
```

10.3 Spare Parts Inventory

Part	Qty	Location	Reorder Level
Vibration sensor PCB 682C09	10	Warehouse	5
Temperature sensor Emerson 644H	10	Warehouse	5
Pressure transmitter Emerson 3051S	5	Warehouse	2
Jetson AGX Orin module	2	Site	1
Industrial switch Cisco IE-4010	1	Site	1
UPS battery pack	5	Warehouse	2
24VDC power supply 10A	5	Warehouse	2
Ethernet cable 10m	20	Site	10
Fiber patch cord 3m	10	Site	5

APPENDIX A: WIRING DIAGRAMS

A.1 Sensor Junction Box

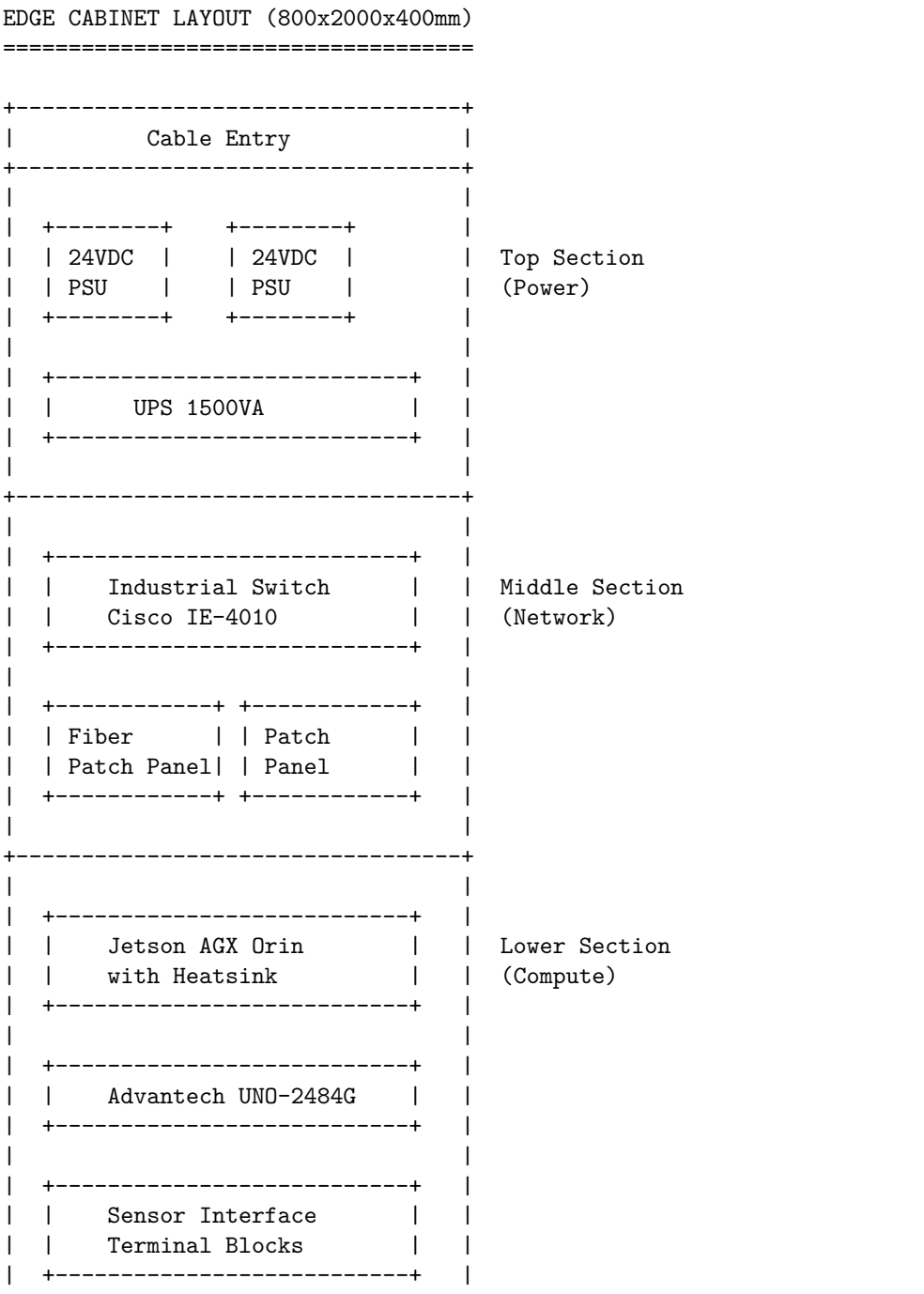
```
SENSOR JUNCTION BOX WIRING
==========================

Terminal Block Layout:
+------+------+------+------+------+------+------+------+
|  1   |  2   |  3   |  4   |  5   |  6   |  7   |  8   |
+------+------+------+------+------+------+------+------+
| VIB+ | VIB- | TEMP+| TEMP-| PRES+| PRES-| 24V+ | 24V- |
+------+------+------+------+------+------+------+------+
   |      |      |      |      |      |      |      |
   |      |      |      |      |      |      +------+----> To Power Supply
   |      |      |      |      +------+----------------> To Edge (AI3)
   |      |      +------+-------------------------------> To Edge (AI2)
```

```
   +------+-----------------------------------------> To Edge (AI1)

Cable Colors (per IEC 60446):
- +24V: Red
- 0V/GND: Blue
- Signal+: White
- Signal-: Brown
- Shield: Green/Yellow
```

A.2 Edge Cabinet Layout



```
|                                   |
+-----------------------------------+
```

APPENDIX B: REFERENCE DOCUMENTS

Document	Number	Revision
P&ID	OG-PID-001	Rev C
Cable Schedule	OG-CAB-001	Rev B
Loop Diagrams	OG-LOOP-001	Rev B
Instrument Index	OG-INST-001	Rev B
Hazardous Area Classification	OG-HAZ-001	Rev A
Earthing Drawing	OG-EARTH-001	Rev A
Network Architecture	OG-NET-001	Rev B

END OF HARDWARE ENGINEER GUIDE